

YI-SHAN LEE

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PROFESSIONAL EXPERIENCE

NVIDIA CORP

Taipei, Taiwan

Senior Digital Design Verification Engineer

Sep 2025 – Present

- Built a Cursor-based automated assertion flow that parsed 100+ interface assertions, cross-referenced Verdi simulation logs, and generated reports, reducing manual review from hours to minutes and enabling 99.2% coverage
- Diagnosed root causes of “No Attempt” assertions by tracing mismatched conditions and undriven signals, coordinated fixes with US DV engineers, and validated changes for next-generation IP reuse
- Shared the Cursor methodology through demos and recordings, enabling 10+ engineers to speed up assertion analysis cycles and accelerate adoption across teams

REALTEK SEMICONDUCTOR CORP. – Top-10 global IC design company

Hsinchu, Taiwan

Digital Design Verification Engineer

Aug 2023 – Aug 2025

- Refined the block-level UVM testbench and functional coverage, built patterns from basic to corner cases, and raised the block to tape-out-ready coverage while taking ownership of later verification stages
- Rebuilt Ethernet SerDes VIP by parameterizing project settings, removing outdated behavior with an empty-DUT setup, and reorganizing test plan and module documentation so future projects could reuse VIP with minimal effort
- Verified SRAM ECC, WDT, and Timer blocks by designing patterns, debugging waveforms, and closing spec gaps, contributing to a clean tape-out and ensuring stable silicon behavior
- Executed chip-level regressions with team scripts, consolidated failure reports into regular status updates, and troubleshoot environment issues such as server or job failures to keep regression results accurate and on schedule
- Integrated the Security VIP by studying standards and IP specs, drafting and presenting a 50-case test plan, developing encryption/decryption patterns, and aligning RTL behavior with protocol requirements

VSLAB, NATIONAL TSING HUA UNIVERSITY

Hsinchu, Taiwan

IT of Server

Sep 2021 – Dec 2022

- Maintained ~7 Linux-based deep learning servers by monitoring system health, resolving crashes and kernel panics, and ensuring stable access for research and training workloads
- Reinstalled OS images, upgraded hardware, configured CUDA/GPU drivers, and standardized training environments to improve reproducibility
- Monitored GPU/CPU utilization, resolved thread-saturation issues, and enforced fair resource allocation
- Managed SSH keys and user accounts and guided researchers through environment debugging to improve workflow
- Trained junior members on server maintenance practices, helping institutionalize lab operating procedures

EDUCATION

NATIONAL TSING HUA UNIVERSITY

Hsinchu, Taiwan

MS in Electrical Engineering, GPA: 3.83/4.30

Sep 2021 – Jun 2023

Advised by Prof. Min Sun. (<http://aliensunmin.github.io/>)

Thesis: *Video and Music Matching via Cross-Modality Lifting* – Creating a content-based system for matching videos and background music, and proposing a cross-modal framework to support the matching process

NATIONAL YANG MING CHIAO TUNG UNIVERSITY

Hsinchu, Taiwan

BS in Electrical and Computer Engineering, GPA 3.33/4.30

Sep 2017 – Jun 2021

Honors: Academic Achievement Award (Top 5% students in the previous semester) (Mar 2018)

PROJECT EXPERIENCE

AN INTERESTING FIGHTING GAME USING COMPUTER VISION – 2-player game that detects real-time actions through a front-facing camera and applies visual effects based on hits

Hsinchu, Taiwan

Developer | National Tsing Hua University | [GitHub](#)

Sep 2022 – Jan 2023

- Designed the core gameplay mechanism incorporating 2-player real-time interaction and facial deformation effects, establishing the technical direction and system logic for the project
- Implemented motion-detection logic using MediaPipe landmarks and geometric rules to recognize punching actions, and built a lightweight 2-player detection pipeline that maintained accuracy and real-time performance on laptops
- Enhanced system robustness by addressing landmark-missing scenarios and supporting stable camera-to-detection-to-rendering integration with teammates

LEARNING-BASED INDOOR LOCALIZATION USING WI-FI CSI – Machine-learning-based indoor localization system using Wi-Fi CSI, noise reduction, and phase calibration for position prediction *Hsinchu, Taiwan*
Developer | National Yang Ming Chiao Tung University | [GitHub](#) Feb 2020 – Jun 2021

- Created the full data-collection and preprocessing pipeline using Intel 5300 CSI data, covering amplitude extraction, noise reduction, normalization, PCA, and model preparation, resulting in two classifiers achieving 0.875 (SVM) and 0.975 (MLP) accuracy
- Analyzed the sensitivity of multiple ML/DL models (SVC, SVR, KNN, NN) to CSI preprocessing choices, and co-developed clustering and classification modules, including DBSCAN, PCA, and SVC with different kernels
- Redesigned the entire preprocessing framework in the final phase by rewriting all code modules, reorganizing the dataset into trainable subcarrier–time-series structures, and implementing a complete evaluation pipeline aligned with prior CSI-based localization research

A GAME FOR SNAKE TOUCHING CUBE – Lightweight C++ game where a player controls a snake that loses length when colliding with blocks *Hsinchu, Taiwan*
Developer | National Yang Ming Chiao Tung University | [GitHub](#) Feb 2019 – Jun 2019

- Defined the gameplay concept and object-oriented structure, including snake movement, collision conditions, and failure logic, forming the basis of the project’s system design
- Implemented core game functions in C++, covering movement updates, collision detection via coordinate comparison, double-buffer–style rendering to reduce flicker, and keyboard-based control
- Resolved visual artifacts, collision errors, and synchronization issues through joint debugging with teammates and integrated all modules into a stable, fully playable game

SKILLS

- **Programming:** Python (Pytorch, NumPy, etc.), C++, HTML/CSS, Verilog/System Verilog/UVM
- **Miscellaneous:** Linux, LATEX(Overleaf), Git, SVN, Verdi, Perforce

PUBLICATIONS

Yi-Shan Lee, Wei-Cheng Tseng, Fu-En Wang, Min Sun “VMCML: Video and Music Matching Via Cross-Modality Lifting”, CVPRW 2024
https://openaccess.thecvf.com/content/CVPR2024W/MULA/html/Lee_VMCML_Video_and_Music_Matching_via_Cross-Modality_Lifting_CVPRW_2024_paper.html

- Developed a cross-modality model for video–music matching and implemented the full training and testing pipeline, including data loading, multi-GPU execution, multi-loss design, RecSys-style ranking metrics, and top-k accuracy evaluation
- Built preprocessing and caching scripts for large-scale datasets and integrated cosine embedding, classification objectives, and margin-based metric learning into a unified framework
- Led all experiments, parameter tuning, and ablation studies, validated result reliability, and authored most of the technical content in the paper and workshop presentation slides